

MAIE5102 Assignment 2

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1 Problem 1: A* Search for the 8-puzzle

1.1 Section 2.1.1: Identify the Inadmissible Heuristic (10 points)

Claim: The Nilsson (1971) heuristic $h(n) = P(n) + 3S(n)$ is **NOT** admissible.

Proof by counter-example. Consider the following state, which is one move away from the goal:

1	2	3		1	2	3
8	4	0		8	0	4
7	6	5		7	6	5

Goal:

The true optimal cost is $h^*(n) = 1$ (slide tile 4 to the right).

Compute $P(n)$ (Manhattan distance). Only tile 4 is displaced: it is at $(1, 1)$ but its goal position is $(1, 2)$. Hence $P(n) = |1 - 1| + |1 - 2| = 1$.

Compute $S(n)$ (sequence score). We check the non-central squares in clockwise order: $(0, 0) \rightarrow (0, 1) \rightarrow (0, 2) \rightarrow (1, 2) \rightarrow (2, 2) \rightarrow (2, 1) \rightarrow (2, 0) \rightarrow (1, 0)$.

The tiles at these positions are: 1, 2, 3, 0(empty), 5, 6, 7, 8. The proper successor sequence in the goal (clockwise) is $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6 \rightarrow 7 \rightarrow 8 \rightarrow 1$.

Position	Tile	Next tile	Proper successor	Score
$(0, 0)$	1	2	2	0 (match)
$(0, 1)$	2	3	3	0 (match)
$(0, 2)$	3	0 (empty)	4	2 (mismatch)
$(1, 2)$	0	—	—	skip (empty)
$(2, 2)$	5	6	6	0 (match)
$(2, 1)$	6	7	7	0 (match)
$(2, 0)$	7	8	8	0 (match)
$(1, 0)$	8	1	1	0 (match)

Center square $(1, 1)$ contains tile 4 (non-empty), so we allot 1. Therefore $S(n) = 2 + 1 = 3$.

Compute $h(n)$.

$$h(n) = P(n) + 3S(n) = 1 + 3 \times 3 = 10.$$

Since $h(n) = 10 > 1 = h^*(n)$, the heuristic **overestimates** the true cost. Hence the Nilsson heuristic is **not admissible**. $\square$

Remark. The other three heuristics are all admissible:

- *Misplaced tiles*: each misplaced tile requires at least one move, so $h_{\text{mis}} \leq h^*$.

- *Manhattan distance*: a classical admissible heuristic since the Manhattan distance for each tile is a lower bound on the moves needed for that tile.
- *Mostow & Prieditis*: for each tile, the indicator of being out-of-row $\leq |\Delta_{\text{row}}|$ and out-of-column $\leq |\Delta_{\text{col}}|$, so $h_{\text{M\&P}} \leq h_{\text{Manhattan}} \leq h^*$.

1.2 Section 2.2: Performance Comparison (20 points)

1. Admissibility

Heuristic	Admissible?	Reason
Misplaced Tiles	Yes	Every misplaced tile needs ≥ 1 move
Manhattan Distance	Yes	Sum of per-tile lower bounds
Nilsson (1971)	No	$3S(n)$ causes over-estimation (see Sec. 1.1)
Mostow & Prieditis	Yes	$h_{\text{M\&P}} \leq h_{\text{Manhattan}} \leq h^*$

2. Efficiency (nodes expanded)

We run A* with the three admissible heuristics on the five provided puzzles and record the number of nodes expanded.

Puzzle	Misplaced Tiles	Manhattan Distance	Mostow & Prieditis
puzzle_1	1747	227	522
puzzle_2	1124	80	264
puzzle_3	2852	152	680
puzzle_4	207	32	67
puzzle_5	12	8	8

Analysis and Conclusions

1. **Dominance relation.** For every state and every tile, we have

$$h_{\text{misplaced}} \leq h_{\text{Mostow\&Prieditis}} \leq h_{\text{Manhattan}}.$$

A heuristic that dominates another one never expands more nodes than the dominated one when used in A*.

2. **Expected efficiency ranking.** Because Manhattan distance $\geq$ Mostow & Prieditis $\geq$ Misplaced Tiles, the number of nodes expanded should satisfy:

$$N_{\text{Manhattan}} \leq N_{\text{Mostow\&Prieditis}} \leq N_{\text{Misplaced}}.$$

The experimental results confirm this prediction across all five puzzles.

3. **Conclusion.** Among the four heuristics, only three are admissible. Manhattan distance is the strongest admissible heuristic and yields the fewest expanded nodes. Mostow & Prieditis lies between Manhattan and Misplaced Tiles both theoretically (dominance) and experimentally (fewer expansions than Misplaced, more than Manhattan). The Nilsson heuristic is inadmissible and therefore cannot guarantee optimal solutions with A*.

2 Problem 2: Constraint Satisfaction Problem (40 points)

The cryptarithmic problem:

$$\begin{array}{r} T \quad W \quad O \\ + \quad T \quad W \quad O \\ \hline F \quad O \quad U \quad R \end{array}$$

2.1 Step 1: List All Constraints (15 points)

Variables: $F, T, W, O, U, R \in \{0, 1, \dots, 9\}$ and carry variables $C_1, C_2, C_3 \in \{0, 1\}$.

Constraints:

- (1) **Column 1 (ones):** $O + O = R + 10 C_1$ (i.e. $2O = R + 10 C_1$)
- (2) **Column 2 (tens):** $W + W + C_1 = U + 10 C_2$ (i.e. $2W + C_1 = U + 10 C_2$)
- (3) **Column 3 (hundreds):** $T + T + C_2 = O + 10 C_3$ (i.e. $2T + C_2 = O + 10 C_3$)
- (4) **Column 4 (thousands):** $C_3 = F$
- (5) **Alldiff:** $\text{Alldiff}(F, T, W, O, U, R)$ (all six letters represent distinct digits)
- (6) **No leading zeros:** $T \neq 0$ (leading digit of TWO)
- (7) **No leading zeros:** $F \neq 0$ (leading digit of FOUR)
- (8) **Carry domains:** $C_1, C_2, C_3 \in \{0, 1\}$

From constraints (4), (7), and (8): $F = C_3 \in \{0, 1\}$ and $F \neq 0$, so $F = 1$ and $C_3 = 1$.

2.2 Step 2: Backtracking Search with FC + MRV + LCV (25 points)

(a) **Initial legal values** (given, before any assignment):

F	T	U	W	R	O
$\{1\}$	$\{2, \dots, 9\}$	$\{0, 2, \dots, 9\}$	$\{0, 2, \dots, 9\}$	$\{0, 2, \dots, 9\}$	$\{0, 2, \dots, 9\}$

Step (b): Assign $F = 1$. MRV selects F (domain size 1).

Infer: $C_3 = 1$. Constraint (3) becomes $2T + C_2 = O + 10$.

Forward checking on constraint (3): For each T value, we need $O = 2T + C_2 - 10$ (with $C_2 \in \{0, 1\}$) to be in O 's domain.

T	Possible O values
2, 3, 4	All negative $\Rightarrow$ prune
5	$\{0, 1\}$; only $O = 0$ in domain $\Rightarrow$ keep
6	$\{2, 3\} \Rightarrow$ keep
7	$\{4, 5\} \Rightarrow$ keep
8	$\{6, 7\} \Rightarrow$ keep
9	$\{8, 9\} \Rightarrow$ keep

Updated domains:

T	U	W	R	O
$\{5, 6, 7, 8, 9\}$	$\{0, 2, \dots, 9\}$	$\{0, 2, \dots, 9\}$	$\{0, 2, \dots, 9\}$	$\{0, 2, \dots, 9\}$

Step (c): Assign $T = 6$. MRV selects T (5 values; all others have 9).

LCV analysis: $T = 5$ fails deeper FC (forces $O = 0, R = 0$, violating Alldiff). For $T \in \{6, 7, 8\}$: each eliminates 10 values from other domains (tied). For $T = 9$: eliminates 11 values. Tie-breaking (smallest number) $\Rightarrow T = 6$.

Constraint (3): $12 + C_2 = O + 10$, so $O = 2 + C_2$, giving $O \in \{2, 3\}$. Alldiff removes 6 from U, W, R .

Updated domains:

O	U	W	R
$\{2, 3\}$	$\{0, 2, 3, 4, 5, 7, 8, 9\}$	$\{0, 2, 3, 4, 5, 7, 8, 9\}$	$\{0, 2, 3, 4, 5, 7, 8, 9\}$

Step (d): Try to assign O . MRV selects O (2 values).

Try $O = 2$ (LCV, smallest first): Infer $C_2 = 0$. From constraint (1): $4 = R + 10C_1$, so $C_1 = 0, R = 4$. Constraint (2): $2W = U$. After Alldiff ($O = 2, R = 4, T = 6, F = 1$): $W \in \{0, 3, 5, 7, 8, 9\}, U \in \{0, 3, 5, 7, 8, 9\}$. FC on $U = 2W$: $W = 0 \rightarrow U = 0$ (Alldiff fail), $W = 3 \rightarrow U = 6$ ($6 \notin \text{domain}$), $W = 5 \rightarrow U = 10$ (invalid), rest invalid. $\Rightarrow W$ **domain empty**.

Failure.

Try $O = 3$: Infer $C_2 = 1$. From constraint (1): $6 = R + 10C_1$, so $C_1 = 0, R = 6$. But $T = 6 = R$: Alldiff violation. $\Rightarrow R$ **domain empty**. **Failure.**

Both values of O fail. **Backtrack to T .**

Step (e): Assign $T = 7$. Next LCV value (tied group: $T \in \{6, 7, 8\}$; $T = 6$ failed, try $T = 7$).

Constraint (3): $14 + C_2 = O + 10$, so $O = 4 + C_2$, giving $O \in \{4, 5\}$. Alldiff removes 7 from U, W, R .

Updated domains:

O	U	W	R
$\{4, 5\}$	$\{0, 2, 3, 4, 5, 6, 8, 9\}$	$\{0, 2, 3, 4, 5, 6, 8, 9\}$	$\{0, 2, 3, 4, 5, 6, 8, 9\}$

Step (f): Assign $O = 4$. MRV selects O (2 values).

LCV: both $O = 4$ and $O = 5$ lead to equal domain reductions (tied at 21 total values eliminated). Tie-breaking: $O = 4$ (smallest).

Infer $C_2 = 0$ (from $14 + 0 = 4 + 10$). Constraint (1): $2 \times 4 = 8 = R + 10C_1$, so $C_1 = 0, R = 8$. Constraint (2): $2W + 0 = U + 0$, i.e. $U = 2W$.

Alldiff ($F=1, T=7, O=4, R=8$): remove $\{1, 4, 7, 8\}$ from W, U .

$W \in \{0, 2, 3, 5, 6, 9\}, U \in \{0, 2, 3, 5, 6, 9\}$.

FC on $U = 2W$:

W	$U = 2W$	Valid?
0	0	Alldiff $W=U$ fail
2	4	$4 \notin U$ domain
3	6	✓
5	10	out of range
6	12	out of range
9	18	out of range

After FC: $W = \{3\}, U = \{6\}, R = \{8\}$.

Step (g): Assign $R = 8$. MRV: R, W, U all have domain size 1. Choose R (alphabetically first). $R = 8$.

Step (h): Assign $U = 6$. $U = 6$.

Step (i): Assign $W = 3$. $W = 3$.

All variables assigned. Search complete.

Solution: $F = 1$, $T = 7$, $W = 3$, $O = 4$, $U = 6$, $R = 8$.

$$\begin{array}{r} 734 \\ + 734 \\ \hline 1468 \end{array}$$

Verification: $734 + 734 = 1468$. ✓

All digits distinct: $\{1, 3, 4, 6, 7, 8\}$. ✓

No leading zeros. ✓